CHAPTER 8 · REFERENCE DEPTH

Frequency Bands & Their Aviation Uses

Chapter 7 told you what a frequency is. This chapter tells you what each slice of the spectrum is used for — and this is where RTR papers harvest most of their marks. Examiners love precise numbers: which band carries ATC voice, which one the glide path, what the transponder replies on, where the emergency frequency sits.

SYLLABUS MAP

Part II (ii) Radio frequency bands · principal aviation frequencies · polarization

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◎ Learning objectives — by the end of this chapter you will be able to…

- Name the radio bands VLF to EHF with their frequency ranges, wavelengths and dominant propagation.
- Explain in depth why aviation uses VHF (AM) for short range and HF (SSB) for long range.
- Recall the principal aviation frequencies and what each does.

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- Explain channel spacing (25 kHz / 8.33 kHz) and calculate channel capacity.
- Define polarization, state the polarization of aviation VHF, and relate antenna length to wavelength.

8.1 The spectrum, band by band

8.2 VHF — the ATC workhorse

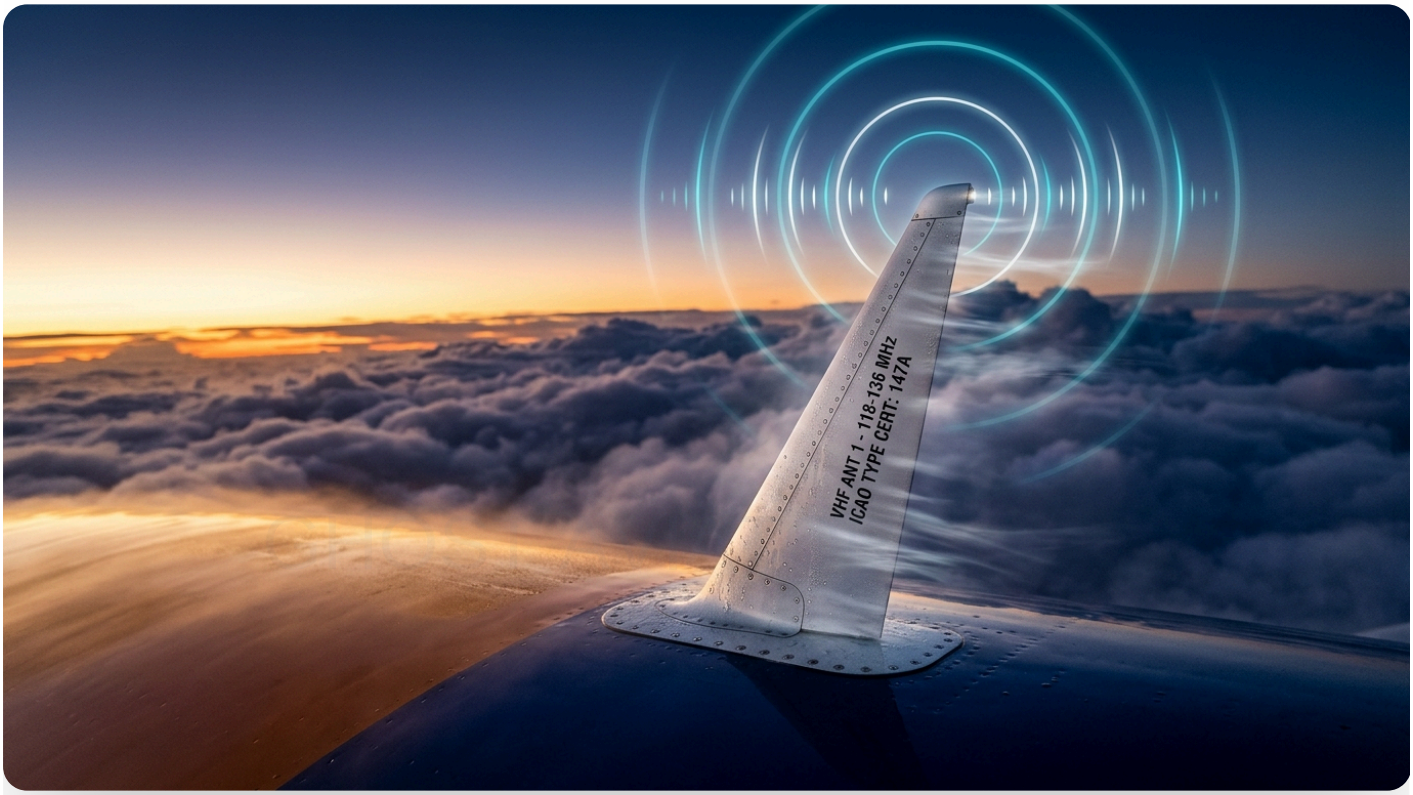
8.3 HF — reaching the ocean

8.4 The aviation frequency master table

8.5 Channel spacing & capacity

8.6 Polarization & antenna length

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The aircraft's antennas are the physical link between the electrical circuits inside and the invisible frequency bands outside.

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8.1 The spectrum, band by band

FIRST PRINCIPLES — WHY THE SPECTRUM IS SPLIT INTO DECADES

The radio spectrum is divided into bands, each a factor of **ten** in frequency above the one below. This matters because the dominant way a wave travels — and therefore what it is useful for — changes with frequency. Low frequencies have long wavelengths and **bend around the Earth**; high frequencies have short wavelengths and travel in **straight lines**. Aviation lives mainly in the **HF, VHF and UHF** bands.

The Radio Spectrum Bands

Band	Frequency	Wavelength	Dominant propagation	Typical aviation use
VLF	3–30 kHz	10–100 km	Ground/wave-guide	(Very long range; rare)
LF	30–300 kHz	1–10 km	Ground wave	NDB (low end)
MF	300 kHz–3 MHz	100 m–1 km	Ground wave	NDB; 2182 kHz maritime distress
HF	3–30 MHz	10–100 m	Sky wave	Long-range / oceanic voice

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Band	Frequency	Wavelength	Dominant propagation	Typical aviation use
				(SSB)
VHF	30–300 MHz	1–10 m	Space (line-of-sight)	ATC voice, VOR, ILS LOC, markers
UHF	300 MHz–3 GHz	0.1–1 m	Line-of-sight	ILS glide path, DME, SSR, GPS, mil voice
SHF	3–30 GHz	1–10 cm	Line-of-sight	Radar, radio altimeter, satcom
EHF	30–300 GHz	1–10 mm	Line-of-sight	(Advanced / experimental)

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MNEMONIC — THE BAND ORDER

"Very Long Modems Help Various Useful Signals Excel" → VLF, LF, MF, HF, VHF, UHF, SHF, EHF. Each band is **×10** the previous.

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8.2 VHF — the ATC workhorse

THE VHF AVIATION COMMUNICATION BAND

118.0 – 136.975 MHz, using **Amplitude Modulation (AM)**, **vertically polarized**, propagating by **line-of-sight (space wave)**. Range is limited to roughly the radio horizon — higher you fly, farther you reach.

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WHY AM AND WHY VHF

VHF gives clear, static-free, short-range communication ideal for an airport's busy frequencies. AM is used so that two overlapping transmissions produce a tell-tale **heterodyne squeal** — alerting everyone that two stations transmitted at once, rather than one silently capturing the channel as FM would (the "capture effect", Chapter 10). AM's "you'll hear the clash" behaviour is a **safety feature**.

8.3 HF — reaching the ocean

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THE HF BAND IN AVIATION

3 – 30 MHz, using **Single Side Band (SSB)** for efficiency, propagating by **sky wave** (refracted off the ionosphere). This bounces signals thousands of miles — essential over oceans and polar routes where VHF line-of-sight fails. Crews change HF frequency between day and night (Chapter 9).

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⚠ EXAM TRAP

Tie the band to its job: **VHF = AM, short-range, line-of-sight; HF = SSB, long-range, sky wave.** Swapping AM/SSB or the propagation path is the classic trap.

8.4 The aviation frequency master table

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Master Frequency Allocations

System / use	Frequency	Band
VHF ATC voice	118.0 – 136.975 MHz	VHF
VHF emergency (guard)	121.5 MHz	VHF
UHF military emergency	243.0 MHz	UHF
SAR scene-of-search (aux)	123.1 MHz	VHF
ELT (satellite / homing)	406 MHz / 121.5 MHz	UHF / VHF

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System / use	Frequency	Band
VOR	108.0 – 117.975 MHz	VHF
ILS Localizer	108.10 – 111.95 MHz	VHF
ILS Glide Path	329 – 335 MHz	UHF
Marker beacons	75 MHz	VHF
DME	962 – 1213 MHz	UHF
SSR transponder	Interrogate 1030 / Reply 1090 MHz	UHF
NDB / ADF	≈ 190 – 1750 kHz	LF/MF
Radio altimeter	4200 – 4400 MHz	SHF
Maritime distress	2182 kHz	MF

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8.5 Channel spacing & capacity

FIRST PRINCIPLES — FITTING MORE AIRCRAFT INTO ONE BAND

The VHF band is a fixed width, so the number of channels it holds depends on how closely they are spaced. Traditionally channels were **25 kHz** apart; to fit more in the crowded European band, **8.33 kHz** spacing was introduced — tripling the number of channels.

WORKED EXAMPLE — CHANNEL CAPACITY

The VHF band is 118.0 to 136.975 MHz, a width of **18.975 MHz = 18,975 kHz.**
At **25 kHz** spacing: 18,975 ÷ 25 ≈ **760 channels**. At **8.33 kHz** spacing: 18,975 ÷ 8.33 ≈ **2,280 channels** — roughly three times as many.
This is why 8.33-spaced frequencies read like **120.275** or **118.905** rather than ending in tidy tenths.

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8.6 Polarization & antenna length

DEFINITION & THE AVIATION FACT

Polarization is the plane of the wave's **electric field — vertical** or **horizontal**. The antenna must match it: a vertical wave needs a vertical antenna. **Aviation VHF communication is vertically polarized**, which is why the COM antenna (the blade) is mounted vertically.

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WHY HIGHER-FREQUENCY AERIALS ARE SHORT

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An efficient antenna is sized to the wavelength — commonly a **half-wave** or **quarter-wave**. Since λ shrinks as frequency rises, VHF aerials are short blades while HF aerials are long wires or probes. From Chapter 7, a 120 MHz λ is 2.5 m, so a quarter-wave whip is about **0.6 m** — a practical blade. An HF λ of 100 m needs a long-wire or a tuned coupler.

✈ COCKPIT REALITY

The little vertical blade on the fuselage is your VHF COM antenna, matched to the vertical polarization of the signal. Mismatch the polarization and received signal strength drops sharply — one reason antennas are mounted exactly as they are.

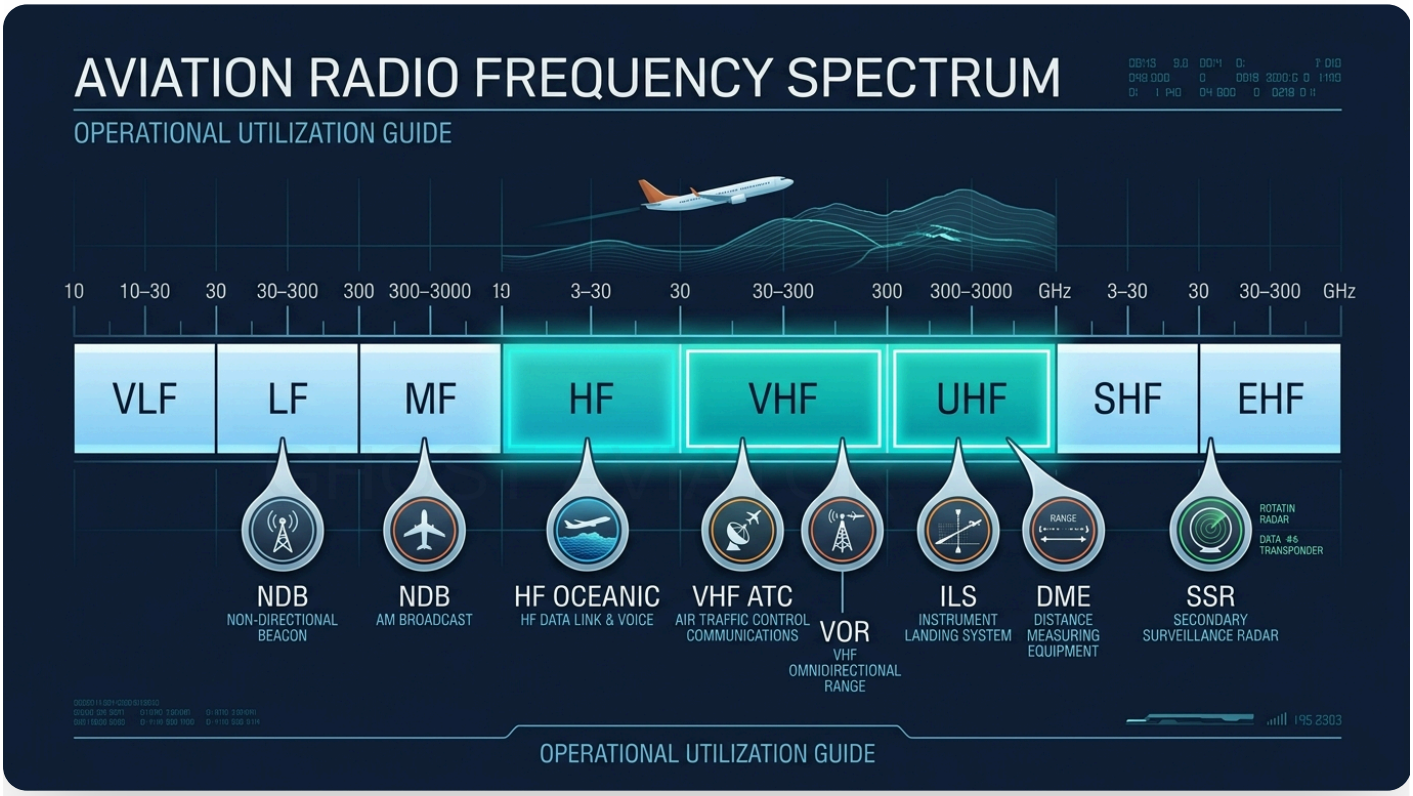


Figure 8.1 — The radio spectrum and where aviation lives.

☆ Numbers to memorise

Essential Facts for Chapter 8	
Fact	**Value**
Band order (×10 each)	VLF · LF · MF · HF · VHF · UHF · SHF · EHF
VHF ATC voice	118.0–136.975 MHz · AM · vertical · line-of-sight
HF voice	3–30 MHz · SSB · sky wave · oceanic
Channel spacing	25 kHz (≈760 ch) and 8.33 kHz (≈2280 ch)
VOR / ILS LOC	108.0–117.975 / 108.10–111.95 MHz (VHF)
ILS GP / DME	329–335 MHz / 962–1213 MHz (UHF)
SSR transponder	1030 MHz interrogate · 1090 MHz reply
Emergency	121.5 (VHF) · 243.0 (UHF) · 406 (ELT) · 123.1 (SAR aux)
Aviation VHF polarization	Vertical

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Question bank

Part A — MCQs (click an option to check)

1. The VHF aeronautical communication band is:

108.0–117.975 MHz

118.0–136.975 MHz

3–30 MHz

329–335 MHz

Answer: 118.0–136.975 MHz.

VHF ATC voice = 118.0–136.975 MHz. 108–117.975 is VOR/ILS-LOC.

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2. Each radio band differs from the next by a factor of:

2

5

10

100

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Answer: 10.

Bands run in decades — each is ten times the previous in frequency.

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3. Aviation VHF communication uses which modulation?

AM

FM

SSB

PCM

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Answer: AM.

VHF voice uses AM; HF long-range uses SSB.

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4. Long-range oceanic voice communication uses:

VHF AM

HF SSB (sky wave)

UHF

VLF

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Answer: HF SSB (sky wave).

HF (3–30 MHz) with SSB propagates by sky wave for long range.

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5. The dominant propagation of NDB (LF/MF) is:

Ground wave

Sky wave

Line-of-sight

Satellite

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Answer: Ground wave.

LF/MF travels mainly by the ground (surface) wave.

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6. An SSR transponder replies on:

1030 MHz

1090 MHz

121.5 MHz

75 MHz

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Answer: 1090 MHz.

Interrogation is 1030 MHz; the reply is 1090 MHz.

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7. The ILS glide path operates in which band?

VHF

UHF (329–335 MHz)

HF

MF

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Answer: UHF (329–335 MHz).

The localizer is VHF; the glide path is UHF (329–335 MHz).

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8. ILS marker beacons transmit on:

108 MHz

75 MHz

329 MHz

1090 MHz

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Answer: 75 MHz.

All markers share a 75 MHz carrier, differing in tone/code.

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9. Moving from 25 kHz to 8.33 kHz channel spacing roughly:

Halves the number of channels

Triples the number of channels

Has no effect

Doubles the band width

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Answer: Triples the number of channels.

8.33 kHz is one-third of 25 kHz, so about three times as many channels fit.

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10. Aviation VHF communication is:

Horizontally polarized

Vertically polarized

Circularly polarized

Unpolarized

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Answer: Vertically polarized.

Aviation VHF comms are vertically polarized — hence the vertical blade antenna.

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11. Higher-frequency antennas are physically shorter because:

They use less power

Wavelength shrinks as frequency rises, and antennas are sized to the wavelength

They are horizontally polarized

They carry less data

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Answer: Wavelength shrinks as frequency rises, and antennas are sized to the wavelength.

Antennas are a fraction of a wavelength; shorter λ at high f means a shorter aerial.

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12. The radio altimeter operates in the:

VHF band

HF band

SHF band (≈4200–4400 MHz)

LF band

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Answer: SHF band (≈4200–4400 MHz).

The radio altimeter works around 4200–4400 MHz in the SHF band.

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13. The maritime distress frequency in the MF band is:

121.5 MHz

2182 kHz

406 MHz

243.0 MHz

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Answer: 2182 kHz.

2182 kHz is the maritime distress frequency (MF).

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14. The international VHF emergency frequency and the UHF military one are:

121.5 / 122.5 MHz

121.5 / 243.0 MHz

118.0 / 136.975 MHz

406 / 121.5 MHz

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Answer: 121.5 / 243.0 MHz.

121.5 MHz (VHF guard) and 243.0 MHz (UHF military guard); 243 = 2 × 121.5.

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15. VOR operates in the:

VHF band (108.0–117.975 MHz)

UHF band

HF band

MF band

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Answer: VHF band (108.0–117.975 MHz).

VOR is VHF, 108.0–117.975 MHz (sharing 108–111.95 with ILS-LOC).

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16. DME operates in the:

VHF band

UHF band (962–1213 MHz)

SHF band

LF band

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Answer: UHF band (962–1213 MHz).

DME is UHF, 962–1213 MHz.

Part B — Oral / viva (tap to reveal model answers)

▼ ? **State the VHF communication band and its characteristics.**

Model Answer:

118.0–136.975 MHz, amplitude modulated, vertically polarized, propagating line-of-sight (space wave) for short range.

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▼ ? **Why is HF used for oceanic communication?**

Model Answer:

HF (3–30 MHz) propagates by sky wave, refracting off the ionosphere to reach thousands of

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miles where VHF line-of-sight cannot. SSB is used for efficiency, and the frequency is changed between day and night.

▼ ? Name the bands and uses for the ILS components.

Model Answer:

Localizer — VHF, 108.10–111.95 MHz (azimuth); glide path — UHF, 329–335 MHz (vertical); marker beacons — 75 MHz (distance).

▼ ? On what frequencies does an SSR transponder interrogate and reply?

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Model Answer:

Ground interrogation on 1030 MHz; airborne reply on 1090 MHz.

▼ ? What is channel spacing, and why was 8.33 kHz introduced?

Model Answer:

The frequency gap between adjacent channels. 8.33 kHz spacing was introduced to fit about three times as many channels into the crowded VHF band as the older 25 kHz spacing.

▼ ? What is the polarization of aviation VHF, and why does it matter?

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Model Answer:

Vertical — the COM antenna is a vertical blade to match it. If the antenna polarization is mismatched, received signal strength drops sharply.

⚡ 60-SECOND REVISION CARD

Bands ×10: **VLF·LF·MF·HF·VHF·UHF·SHF·EHF**; aviation core = HF/VHF/UHF.

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VHF 118–136.975 MHz, AM, vertical, line-of-sight; HF 3–30 MHz, SSB, sky wave, oceanic.
ILS: LOC **VHF** 108.10–111.95 · GP **UHF** 329–335 · markers **75 MHz**. DME **UHF** 962–1213.
SSR **1030/1090**.
Emergency **121.5 / 243.0 / 406**, SAR aux **123.1**. Channel spacing **25 / 8.33 kHz** (≈760 / 2280 ch).
Aviation VHF **vertically polarized**; antenna length scales with wavelength.

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